Reg. No. [ ] [18AI5DE12]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

SPEECH PROCESSING

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Explain how operators was trained to make the voder talk.	**08**
	b.	Compare von kemplelen's speaking machine with Dudley's voder.	**06**
2.	**a.**	Prove that if y(n) is the convolution of two sequences x(n) and h(n), then the z transform Y(z) is the product of the z transforms X(z) and H(z).	**08**
	b.	Explain in detail about the components of automatic speech recognition with the help of block diagram	**06**
3.	**a.**	Establish the wave Equation for Vibrating String with suitable pictures	**08**
	b.	As the mass density in a tube increase, does the tube impedance increase, decrease, or stay the same? Explain	**06**
		OR	
4.	**a.**	Consider an acoustic tube excited at one end by the complex exponential and open at the other end. Prove that the volume velocity in the tube directly proportional to pressure gradient.	**08**
	b.	Sketch an acoustic tube model of a voiceless fricative sound such as /sh/. This sketch should be of a qualitative nature accompanied by some intuitive justification of your solution.	**06**
5.	**a.**	Explain with diagrams how adaptation and tuning takes place in the human ear.	**07**
	b.	State Von Helmholtz's place theory. Who proved it wrong and how?	**07**
		OR	
6.	**a.**	What is the difference between place model periodicity and model of pitch perception?	**07**
	b.	The auditory system behaves like some sort of a filter bank. Explain this statement with reference to Fletcher's critical ratio experiment.	**07**
7.	**a.**	Show that linear prediction can be used to generate estimates of the spectral envelope for the short-term spectrum of speech that is commonly of interest for applications in synthesis and recognition	**07**
	b.	What are the features that can be used to categorize the sounds on any language?	**07**

OR

8.	**a.**	Prove that complex cepstrum, If $\hat{x}(n) = 0$ for $n > 0$, this must correspond to an all-zero (FIR) filter with all the zeros outside the unit circle.	**07**
	b.	Discuss about statistical modeling for speech generation.	**07**